

Practice Test Answer Key

Total Score: 10 / 15

Correct Answers: 10 / 15

1. Which of the following is an example of a non-linear data structure?

1. Array
2. Stack
3. Queue
4. Graph

(Your Selection)

Correct Answer: Option 4

Your Answer: Option 4

2. A hash function maps keys to:

1. Values
2. Memory locations
3. Indices in a hash table
4. Data types

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 1

3. Which of the following algorithms is used to find the shortest path in a weighted graph?

1. Kruskal's algorithm
2. Prim's algorithm
3. Dijkstra's algorithm
4. DFS

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 3

4. Which of the following is NOT a linear data structure?

1. Array
2. Linked List
3. Graph
4. Queue

(Your Selection)

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 3

5. In a binary tree, each node can have at most:

1. 1 child
2. 2 children
3. 3 children
4. Unlimited children

(Your Selection)

Correct Answer: Option 2

Your Answer: Option 2

6. Which algorithm is best for finding a minimum spanning tree?

1. DFS
2. BFS
3. Kruskal's Algorithm
4. Bellman-Ford

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 1

7. Which is the correct time complexity of Bubble Sort in the worst case?

1. $O(n)$
2. $O(n \log n)$
3. $O(n^2)$
4. $O(\log n)$

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 3

8. Which of the following data structures allows insertion and deletion at both ends?

1. Queue
2. Deque
3. Stack
4. Priority Queue

(Your Selection)

Correct Answer: Option 2

Your Answer: Option 2

9. Which data structure is best suited for recursive function calls?

1. Stack
2. Queue
3. Heap
4. Tree

(Your Selection)

Correct Answer: Option 1

Your Answer: Option 4

10. Which data structure uses LIFO (Last In First Out) order?

1. Queue
2. Stack
3. Array
4. Linked List

(Your Selection)

Correct Answer: Option 2

Your Answer: Option 2

11. Time complexity of searching in a balanced Binary Search Tree (BST):

1. $O(1)$
2. $O(\log n)$
3. $O(n)$
4. $O(n \log n)$

(Your Selection)

Correct Answer: Option 2

Your Answer: Option 2

12. Which of the following has the fastest average time complexity for searching?

1. Linked List
2. Stack
3. Hash Table
4. Queue

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 3

13. Which one is not a stable sorting algorithm?

1. Merge Sort
2. Insertion Sort
3. Selection Sort
4. Bubble Sort

(Your Selection)

(Your Selection)

Correct Answer: Option 3

Your Answer: Option 4

14. Which of the following correctly arranges the time complexities from fastest to slowest (as n becomes large)

1. $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n)$
2. $O(\log n) < O(1) < O(n) < O(n \log n) < O(n^2) < O(2^n)$
3. $O(1) < O(n) < O(\log n) < O(n \log n) < O(2^n) < O(n^2)$
4. $O(1) < O(n \log n) < O(\log n) < O(n) < O(2^n) < O(n^2)$

Correct Answer: Option 1

Your Answer: Option 1

15. Which data structure uses FIFO (First In First Out) ordering?

1. Stack
2. Queue
3. Array
4. Tree

(Your Selection)

Correct Answer: Option 2

Your Answer: Option 2